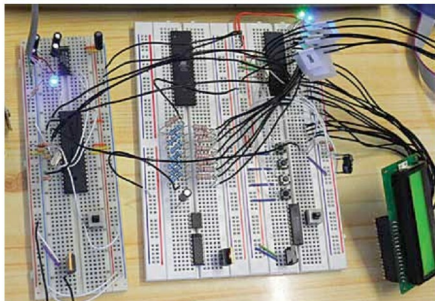


Electronic Circuit Prototyping More Than Just An Artefact

Certain doubts in the design of an electronic product can be cleared by building and studying a prototype first. Here we discuss the challenges and different approaches to prototyping



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Prototypes are not just a vital element of the electronics design but a powerful tool for public communication and exhibition too. Creating a physical mock-up of an object makes it possible to understand the ways people will attempt to interact with it, and the ways it can be manipulated. Concisely, in the electronics design and development process, prototypes can be used for different purposes including experimentation, learning, testing, proofing, communication and interaction.

Prototype wiring

Today, designers and DIY practitioners are using various prototyping ideas to express their concepts of electronic ar-

tefacts. These prototypes are generally constructed using various solder and/or solder-free methods. All of these methods have their own pros and cons:

Wire-wrapping and point-to-point wiring. In wire-wrapping method, fine wire is tightly connected to endpoints

by wrapping onto a rectangular pin—instead of soldering—with the help of a special gun tool.

In point-to-point wiring, components are soldered to a perforated board, also known as Vero board.

Both wire wrapping and point-to-point methods are inexpensive, but wire wrapping allows reliable connections to square leads only. So it becomes necessary to solder discrete components to a header that goes into an IC socket or to individual wire-wrap pin. With point-to-point wiring, it is easy to simply twist, solder leads and run wires where needed. However, this method is slow and requires ample skill.

Solderless wiring. The invention of the solderless breadboard—an improvement over the basic wire-wrapping and point-to-point wiring—was a significant step over many of the other prototyping methods. Breadboard is a quick way to set up any circuit using standard through-hole components. But breadboards are not good for high-frequency circuits due to stray capacitance and inductance. Further, these are not suit-

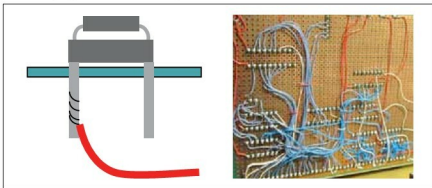


Fig. 1: Wire wrapping and point-to-point wiring